Spring-2026

Statistics for Data Science (CSE303)

Project Title:

Logistics of Relief: A Statistical Analysis of Gaza Aid Supply & Dispatch Patterns

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Introduction:

The main objective of this project is to implement statistical methods and computational tools to solve real-world data science problems using an original dataset. In this regard, we have chosen a dataset named *"State of Palestine: Gaza Supplies and Dispatch Tracking."* The dataset was taken from the Humanitarian Data Exchange (HDX), managed by the United Nations Office for the Coordination of Humanitarian Affairs (OCHA).

This dataset primarily explains from where the relief supplies came from, how many trucks were launched to the war-torn nation of Palestine, and what kinds of products were delivered (such as food items, tents, hygiene kits, etc.).

We, as a team, have studied the records and went through the data and tried to figure out the hidden decisions behind the entries. First of all, we have cleaned the dataset and then observed if there are any null values or duplicacies that exist or not. Solved those issues and figured out which kinds of data are mostly available? Numerical or categorical? Then began using Python's statistical tools.

Exploratory Data Analysis:

We first created a copy of the entire dataset using *copy()* function so that any accidental changes would not affect the original dataset. After that, we used the *head(10)* and *tail(10)* functions to display the first and last 10 datapoints of a dataset. To view information such as datatypes and memory usage, we used the info*()* function. To summarize the dataset such as minimum, maximum, median and standard deviation we used *info()* and *describe()* functions.

The first step is to clean the data by identifying missing values, duplicacies and performing sanity checks. Without proper data cleaning, statistical analysis can be biased, and data counts and averages may become inflated. Moreover, it can lead to wrong business decisions and mislead machine learning models. That is why we first checked for null values using the *isnull()* function. If any can be found, then we can replace it with mean or median values. In our case, null values were found in the categorical columns of the dataset.

We replaced them with the word 'Unknown' by using the *fillna()* function. We did not remove the duplicate values. As there were some values and those were dependent on various instances for each category.

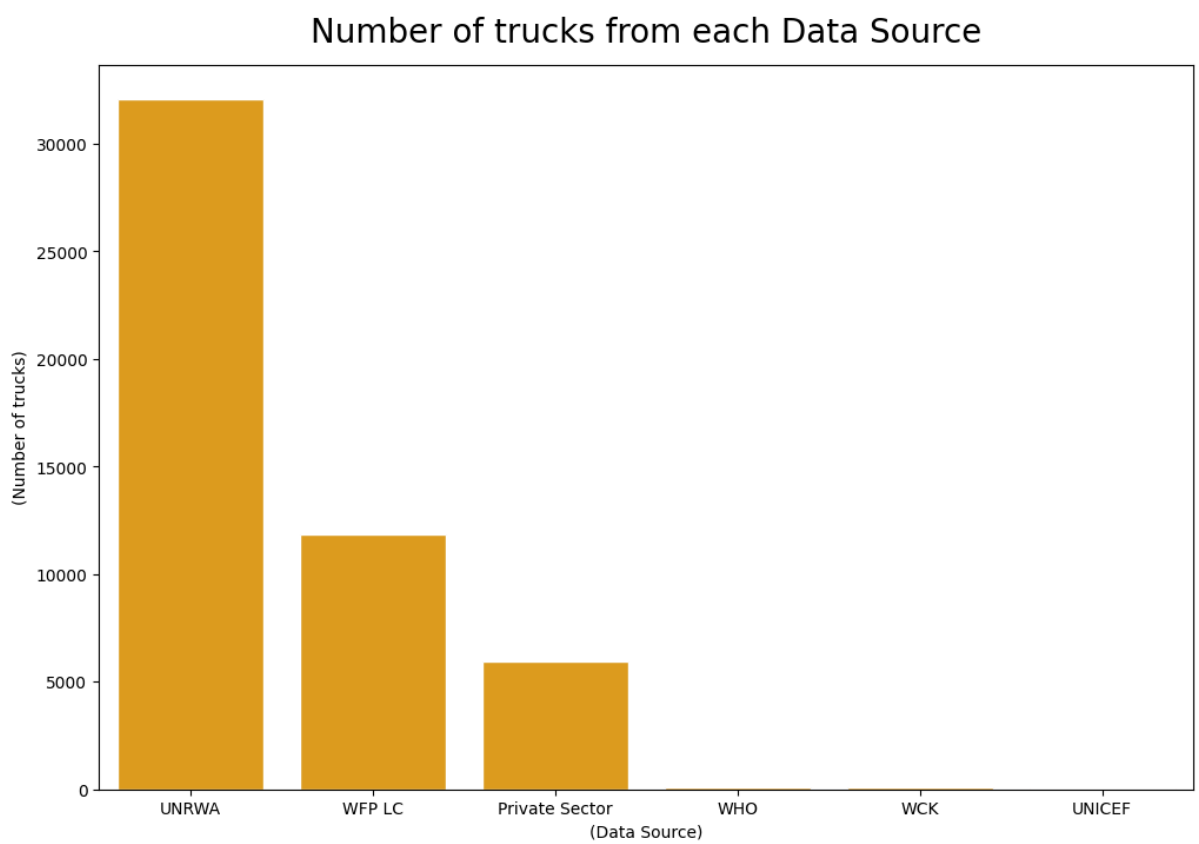
Count Plot:

One approach for visualizing categorical data is a count plot. By count plot, we can witness the frequencies of different categories. From our dataset, we have observed the number of trucks from each data source, the number of top 15 cargo kinds, the variety of donations, and finally, the truckload of each crossing point.

The *countplot()* function from the Python data visualization module Seaborn was used to create the diagrams.

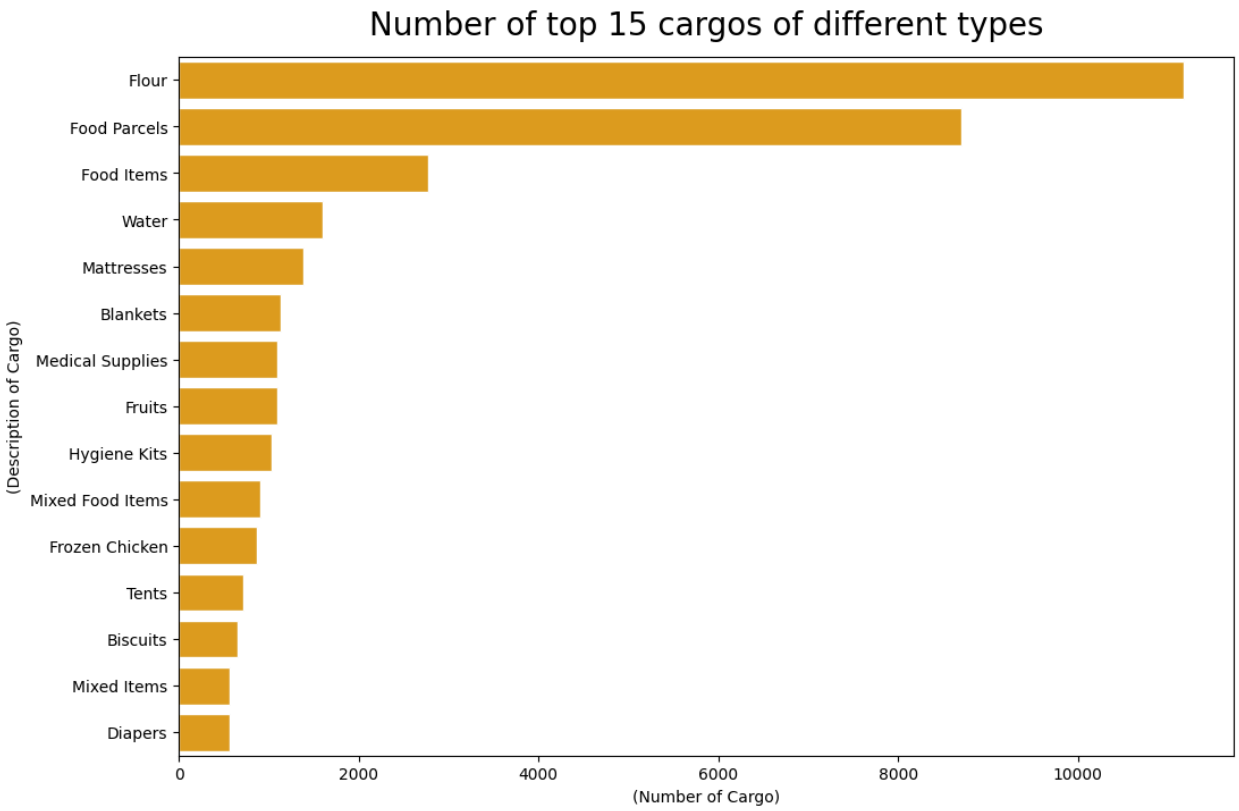
The schematics are as follows:

1.



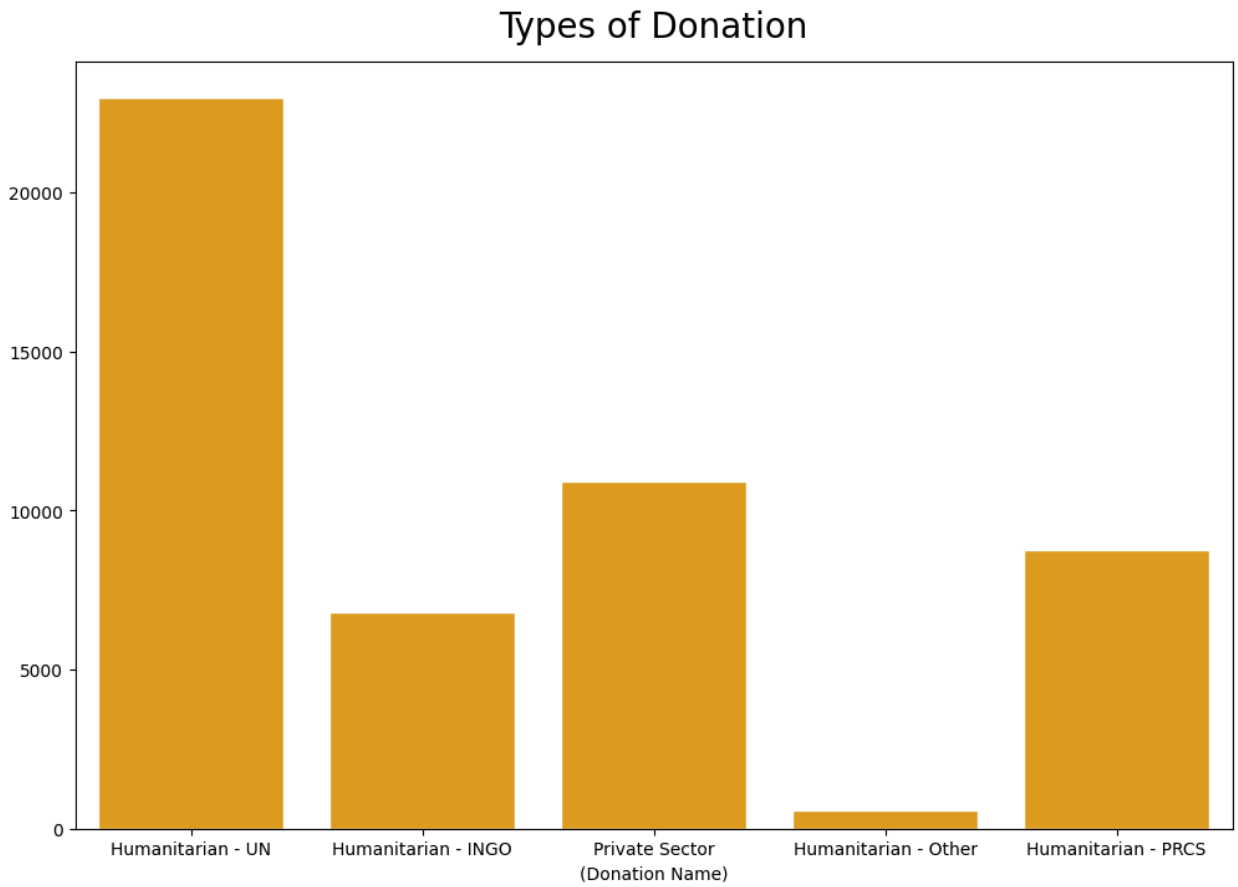
The aforementioned figure shows that the maximum number of goods were served by the UNRWA, whereas the WHO and UNICEF have zero contribution. However, the private sector had contributed with more than 5000 trucks.

2.



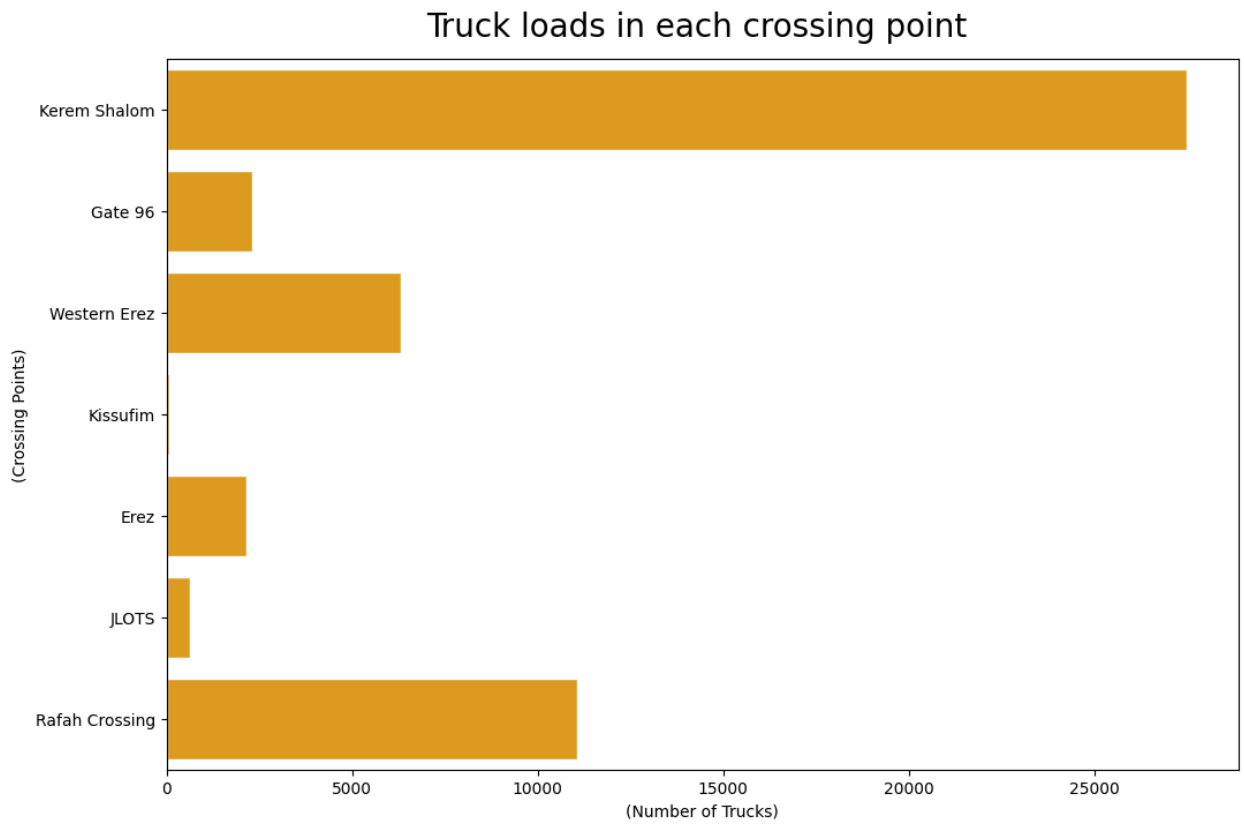
This figure demonstrates that the food items are on top as the relief goods. Besides, it demonstrates that the major meal in Gaza is made with flour. Baby diapers and hygiene kits are also present as the priority items.

3.



The main one, the United Nations, issued about 20,000 humanitarian gifts. In addition, the private sector was just behind them with almost 50% less contribution. Additional donations came from PRCS and INGO.

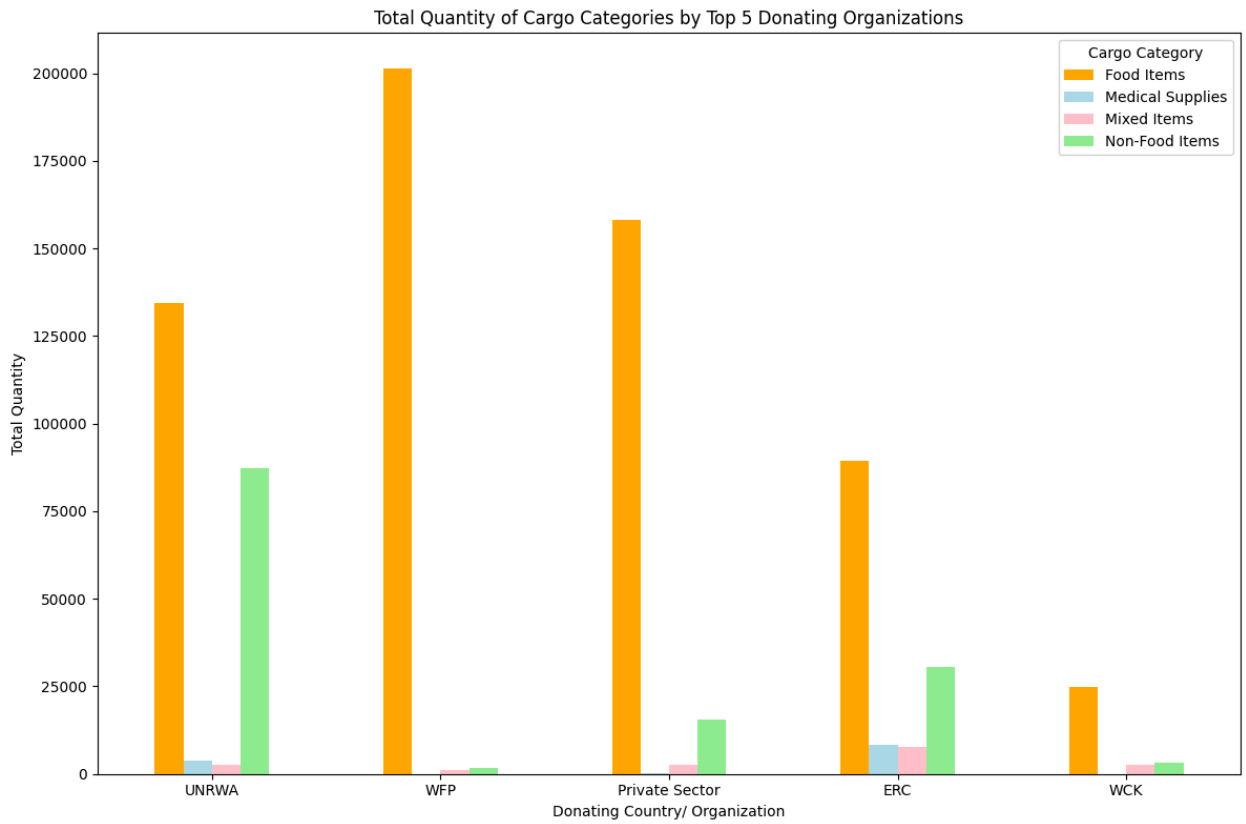
4.



The Kerem Shalom and Rafah crossings are the main entrances for supplying relief in Palestine. On the contrary, Kissufim was not utilized at all.

Bar Plot:

A bar plot represents distinct categories, where the length of each bar signifies the magnitude. It helps us to compare higher or lower values across different categories. Here, we created a bar plot of Donating Country/Organisation by Quantity for each Cargo Category. We used only the top 5 donating organisations for better insight.



The bar chart shows that WFP provides the highest amount of food items, nearly 200,000 units. The Private Sector has around 150,000 units and UNRWA with approximately 130,000 units. On the other hand, UNRWA contributes a significant amount of non-food items, nearly 85,000 units. The ERC and Private Sector provide moderate levels of non-food items. However, WFP and WCK have the lowest contributions across the categories.

Pie Chart:

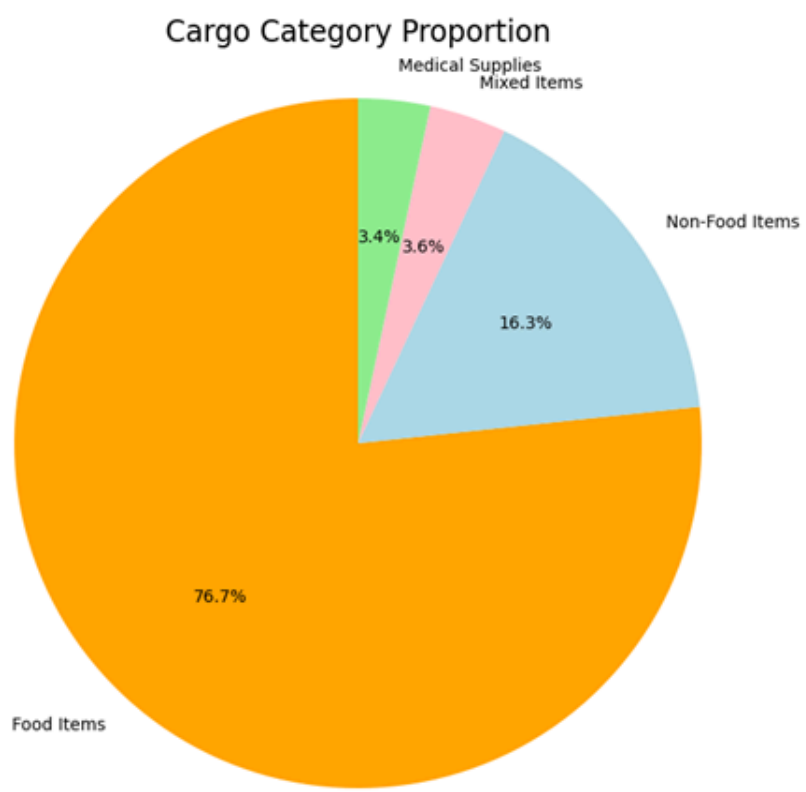
Another useful method for visualizing categorical data is a pie chart. A pie chart represents data in a circular form where each slice shows the proportion of a category compared to the whole dataset. It helped us understand the relative distribution of different categories easily.

In this project, pie charts were used to analyze the proportional distribution of:

- Cargo categories
- Donation types
- Data sources
- Crossing points

Before creating the pie charts, we ensured that pie charts were used only for columns with a limited number of categories (7 or fewer) to maintain clarity and readability.

1. <u>Cargo Category Distribution</u>:

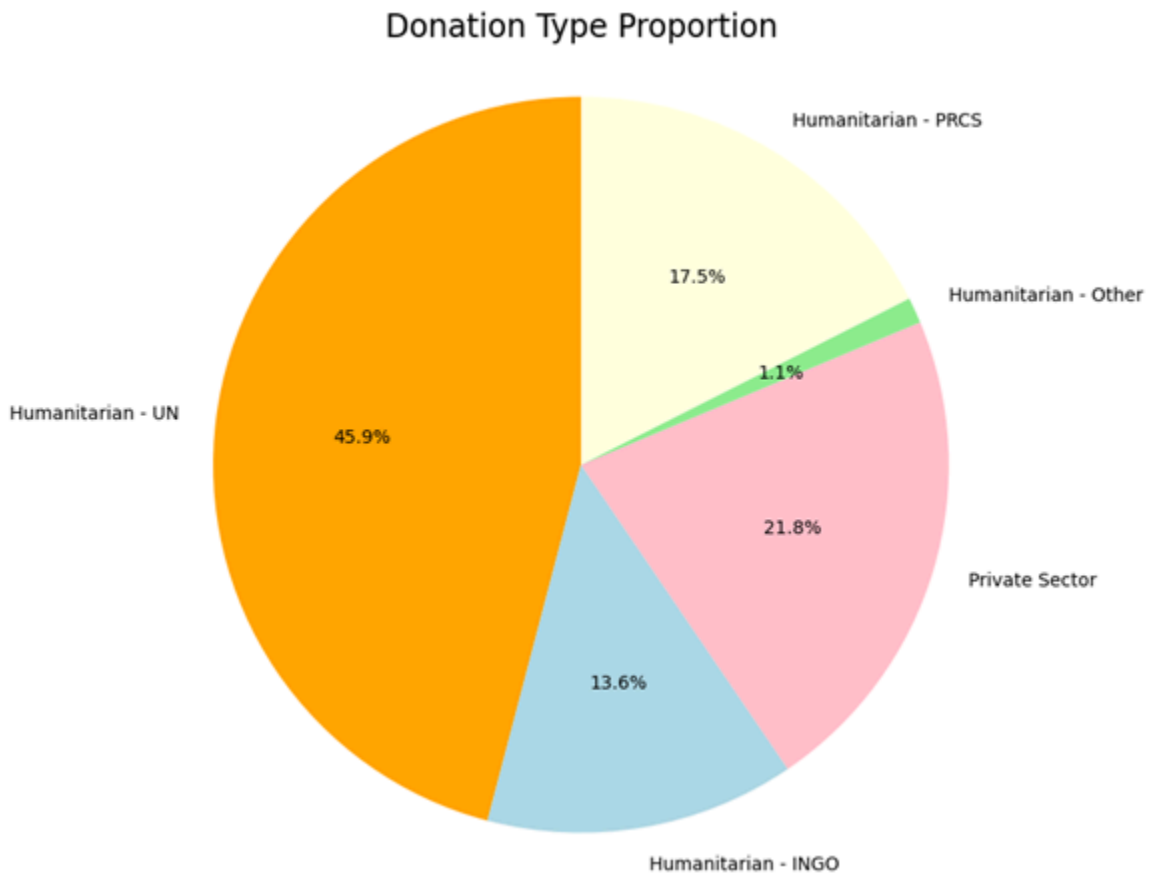


The pie chart for cargo categories shows that food items dominate the dataset, covering 76.7% of the total shipments. This indicates that the primary focus of humanitarian aid is on providing food support.

Non-food items make up 16.3%, including essential supplies such as tents, blankets, and hygiene kits. Mixed items account for 3.6%, while medical supplies represent 3.4%.

This distribution highlights that food-related aid is the largest component, emphasizing the focus on nutritional support in Gaza.

2. <u>Donation Type Distribution</u>:

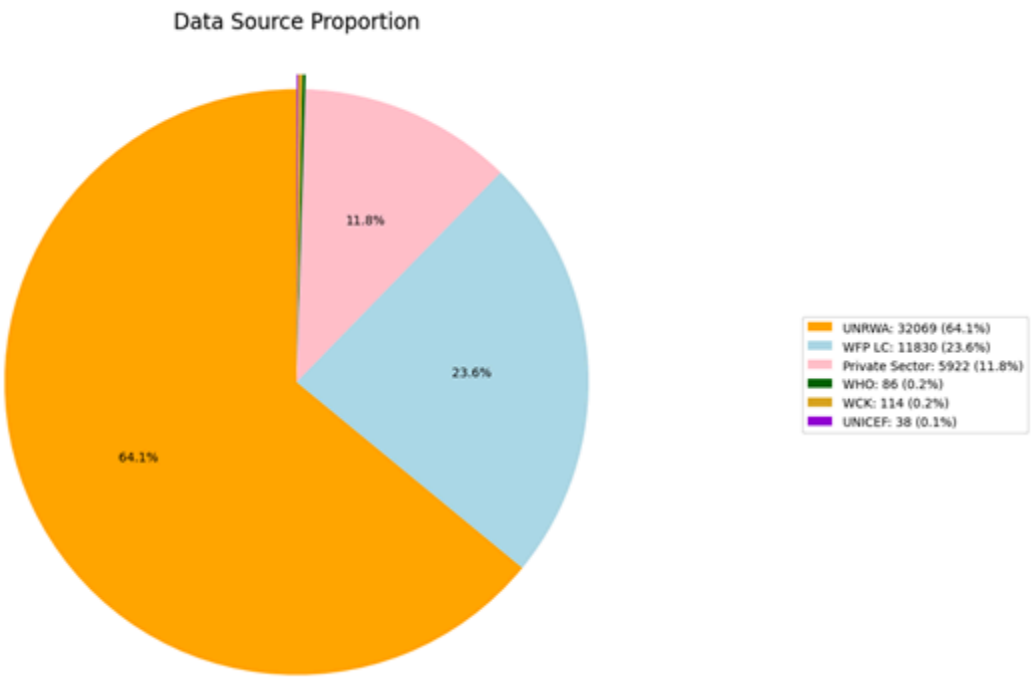


The pie chart for donation types explains that the United Nations is the largest contributor, providing 45.9% of total aid. International NGOs (INGOs) contribute 13.6%, showing their involvement in humanitarian efforts.

The private sector contributes 21.8%, reflecting support from companies and individual donors. PRCS contributes 17.5%, and other organizations provide 1.1% of aid.

The analysis shows that the United Nations leads the humanitarian response, supported by INGOs, the private sector, and PRCS, forming the backbone of aid operations.

3. <u>Data Source Distribution</u>:

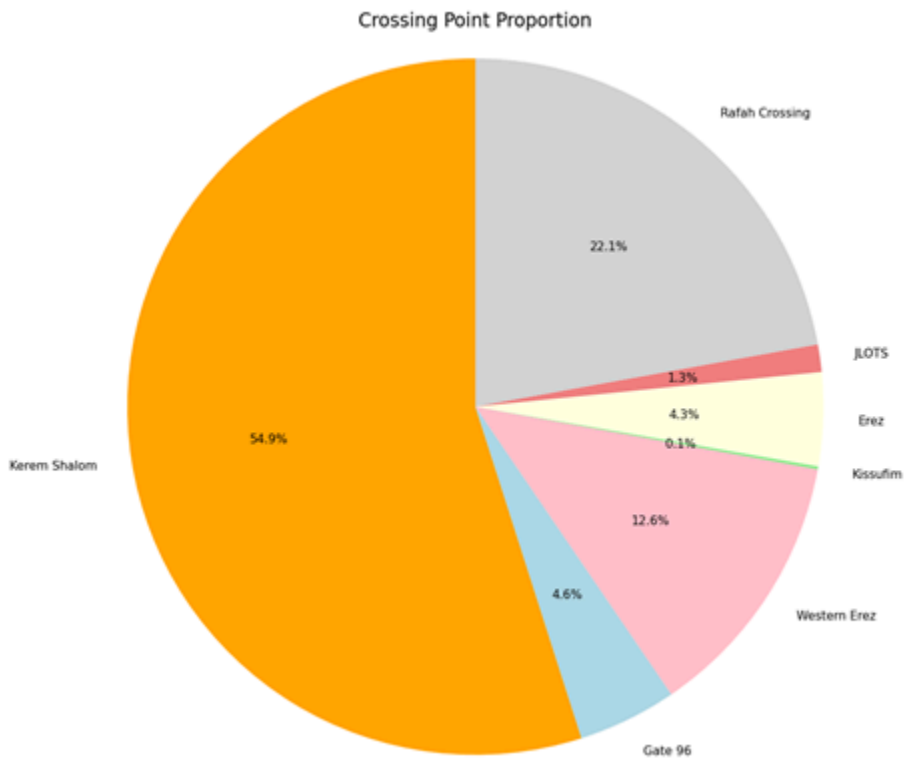


The pie chart for data sources shows a clear dominance of UNRWA, which accounts for 64.1% of total records. This means that most of the data in the dataset comes from this source. The World Food Programme (WFP) contributes 23.6%, making it the second largest source of data.

The private sector provides 11.8% of records. Other sources such as WHO, WCK, and UNICEF have very small contributions (0.2%, 0.2%, and 0.1% respectively).

Overall, UNRWA, WFP, and the private sector together provide the majority of recorded data, while smaller contributors like WHO, WCK, and UNICEF have a minimal share.

4. <u>Crossing Point Distribution</u>:



The pie chart for crossing points shows that Kerem Shalom and Rafah are the main entry points, handling 77% of total aid shipments combined.

Kerem Shalom alone accounts for 54.9%, while Rafah handles 22.1%. Gate 96 contributes 12.6%, Western Erez 4.6%, Erez 4.3%, and JLOTS 1.3%. Kissufim has only a 0.1% contribution.

This indicates that humanitarian aid is highly concentrated through Kerem Shalom and Rafah, which act as critical gateways for relief deliveries.

Excluded Attributes for Pie Charts:

Some columns were not suitable for pie charts due to having too many categories. Destination Recipient/Partner had 9 categories, and Description of Cargo had more than 25 categories.

Using pie charts for these would make the visualization cluttered and difficult to understand. Therefore, these were analyzed using other plots instead, which provide a clearer view of the data.

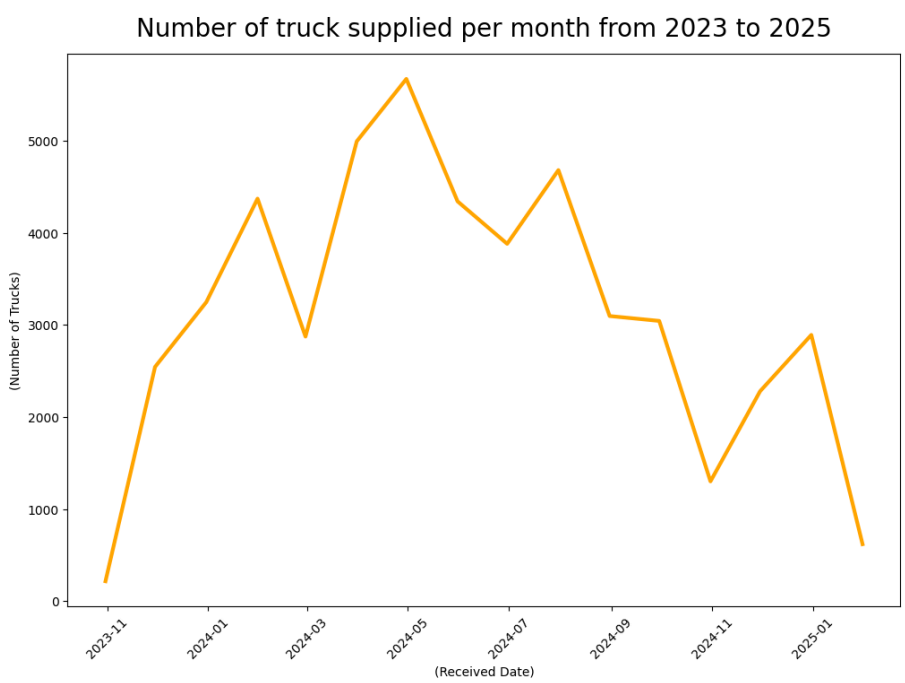
Line Diagram:

A line diagram is used to visualize trends, changes, and relationships, particularly over time. From 2023 to 2025, we have shown the view of the trucks supplied in each month. Moreover, we showed the number of cargo shipments by month during those years. Furthermore, how different crossings have been used over time.

To proceed, we have used the *lineplot()* and *histplot()* functions of the Python Seaborn library.

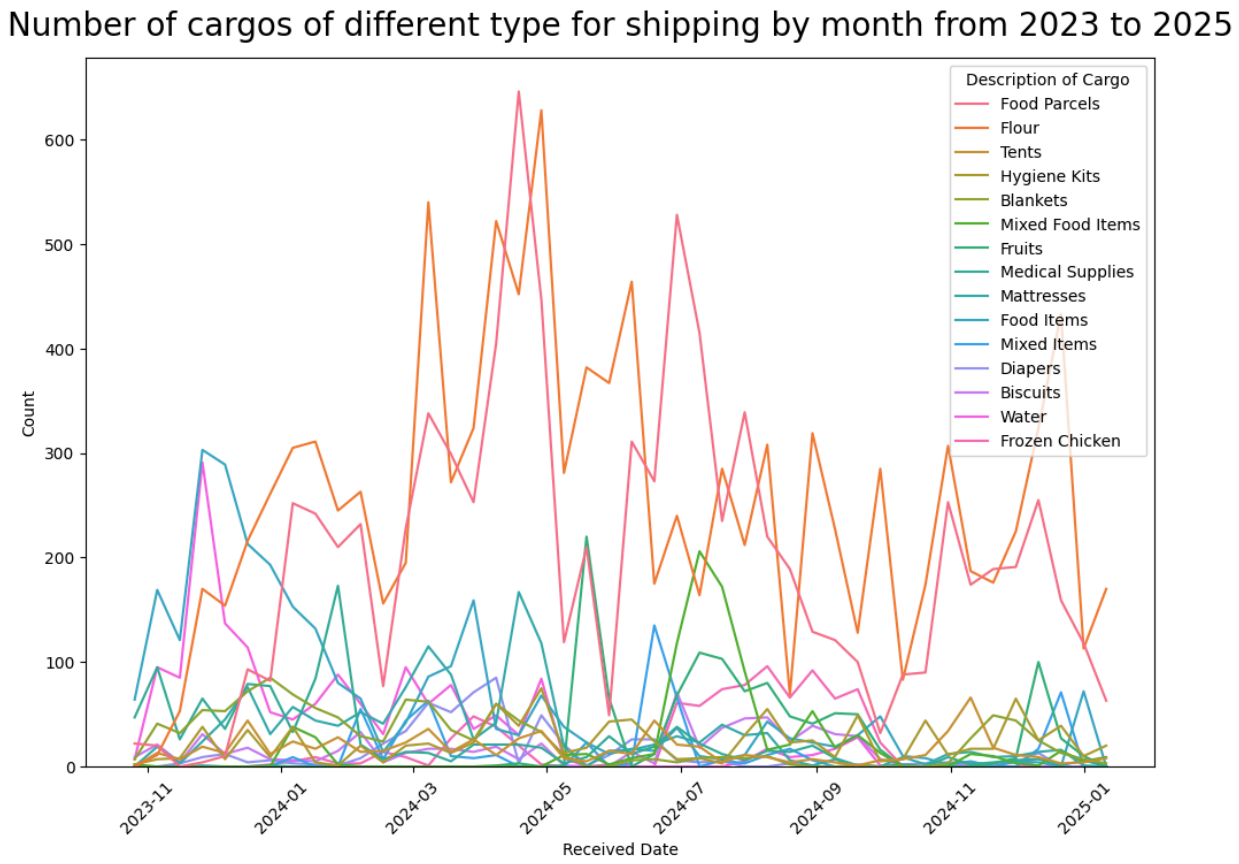
The schematics are as follows:

1.



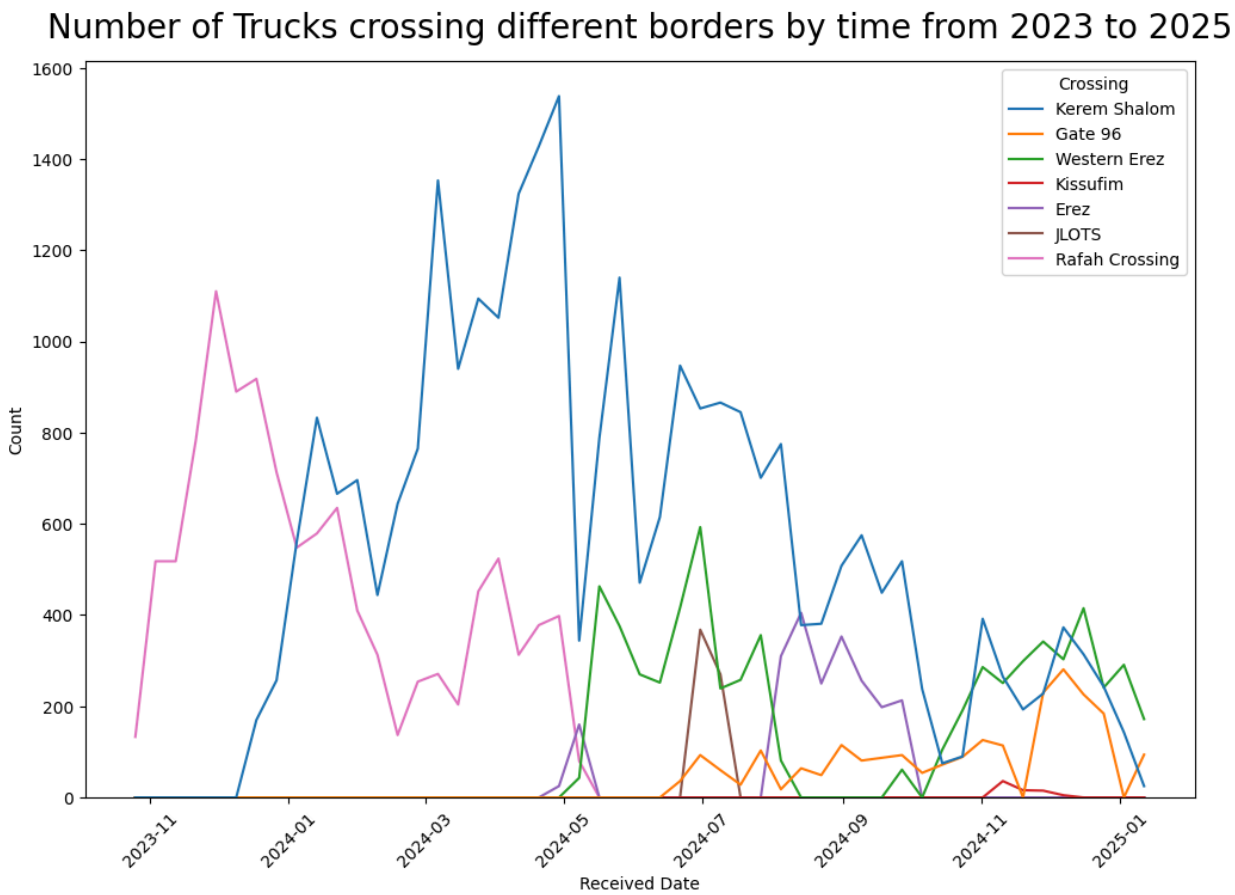
The line diagram shows that, in each span of 2 months from 2023 to 2025, there are ups and downs in the curve. Beginning in November of ’23, the supply of trucks climbed progressively before abruptly declining once it reached nearly 4200. The supply of trucks peaked in May and June of 2024.

2.



The figure demonstrates a scribbled look, as there are 15 different types of cargo shown by month. April and May of 2024 were selected as the months with the greatest availability of food commodities.

3.



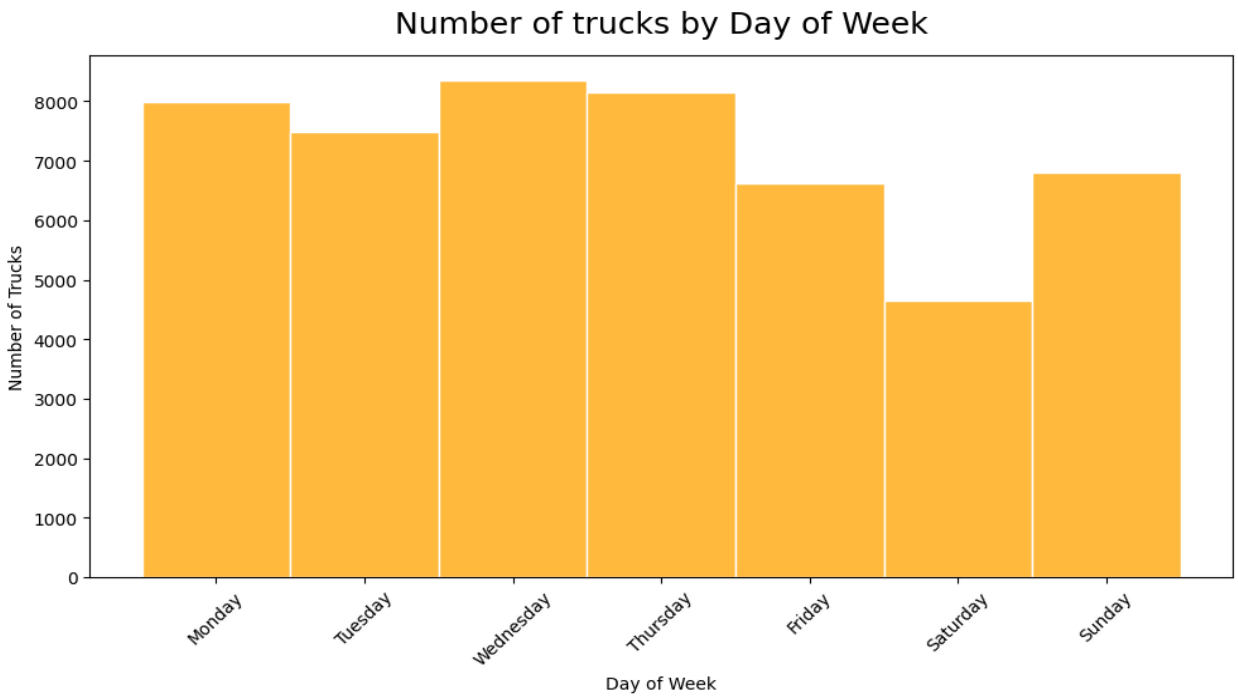
Kerem Shalom was the most frequently utilized crossing for shipping. However, in the first 2 months, it was utilized for none! Later, in April 2024, more than 1500 trucks crossed the gates of Kerem Shalom. Additionally, we can see that the second most used crossing, Gate 96, was never used again after May of '24! There could be any justification to put that off. That might be any serious diplomatic issue or just not used for their own benefit. We can comprehend the other gates' analysis from this graph as the legends are clearly visible.

Histogram:

Histogram is a graphical representation of distribution of continuous numerical data . It divides the data into various ranges which is called interval and shows a bar for each interval. The height of the bar often represents the frequency of data points that comes into the interval .

Histogram bars touch each other, which very significantly shows how the variables are continuous unlike bar charts .

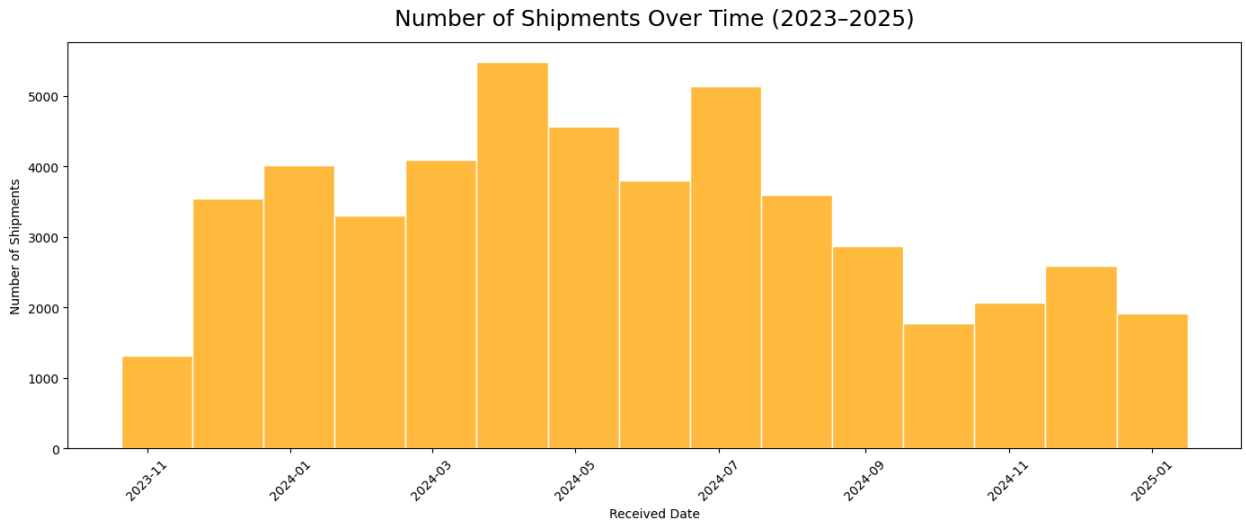
1. <u>Weekly Truck Shipments and Supply Chain Efficiency</u>:



This Histogram shows the total number of trucks received on each particular day of the week from Monday to Sunday .

The illustration indicates that Wednesday is the busiest day of the week, with Monday, Tuesday, and Thursday coming close to exceeding 6000 shipments; however, the remaining days fall short of that, with Saturday falling short of 5000.To maintain a balanced logistics flow, more vehicles are required from Friday to Saturday. Commodities would be distributed more effectively and without needless delays thanks to this equilibrium. This would lessen the needless spread of hunger.

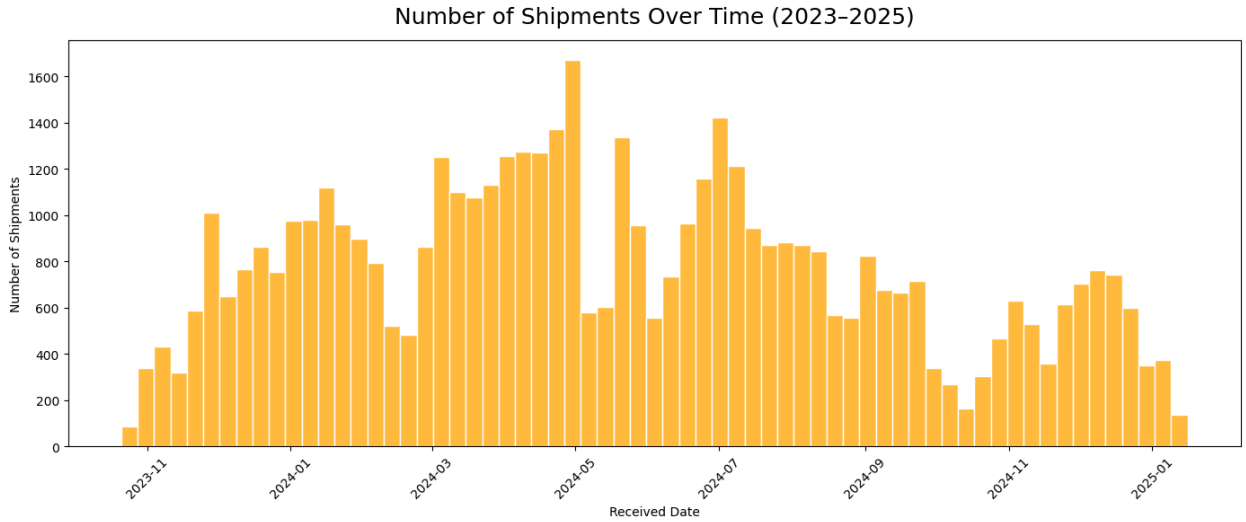
2. <u>Monthly Truck Arrival Volume(2023-2025)</u>:



This histogram aggregates shipments into 30 day bins.This reveals broader monthly trends. The height of each bar corresponds to the total number of trucks received during that month.

The entire situation is visible from a monthly perspective. Prior to Rafah Operation, the number of trucks climbed from over 1,000 to over 5,000 each month; however, following Rafah Operation, the number of trucks gradually decreased, reaching its lowest point in October 2024. which was done on purpose to induce an artificial famine among Gaza's citizens. The monthly view makes it simple to see which month the number of trucks fluctuates and to determine the reason.

3. <u>Weekly Truck Arrival Volume(2023-2025)</u>:

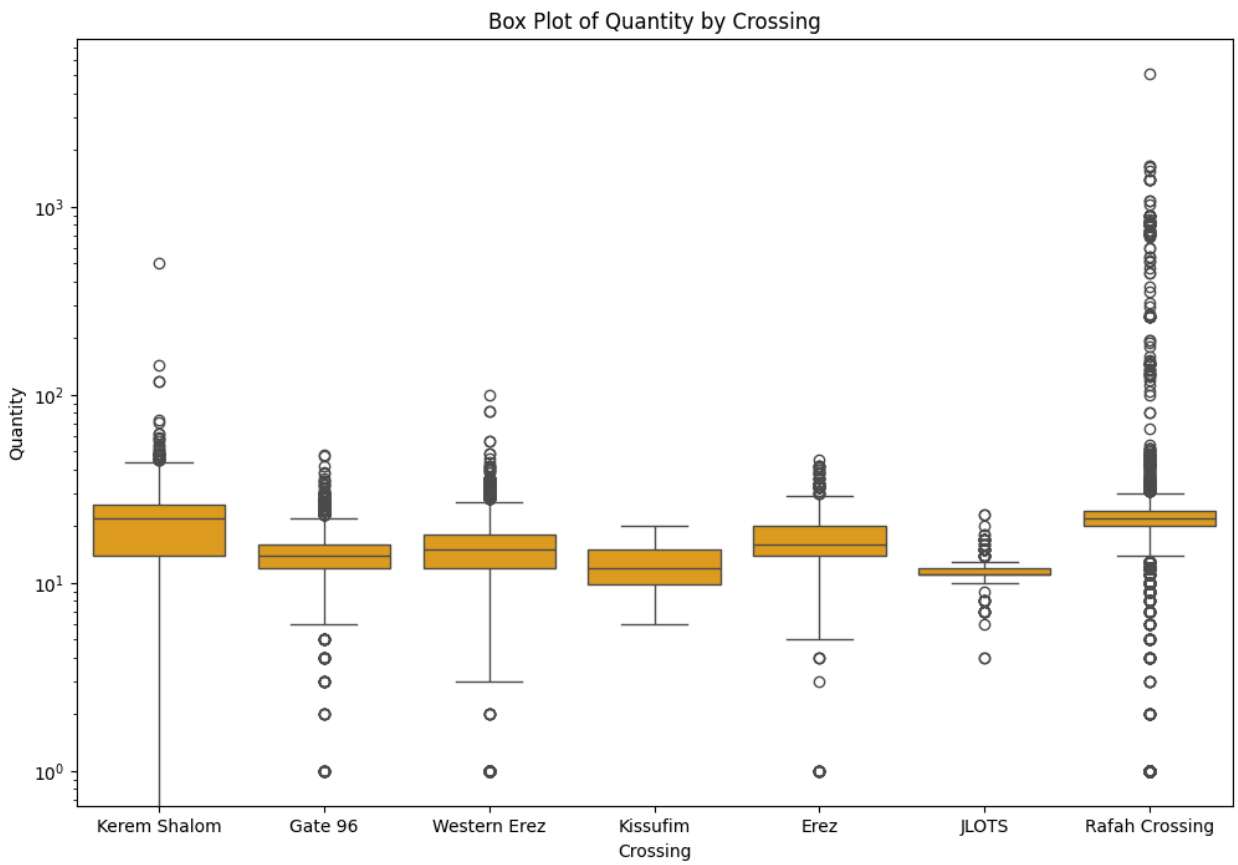


The quantity of shipments received per week between late 2023 and early 2025 is seen in this histogram. The height of each vertical bar shows the total number of shipments that arrived over a seven-day period.

In addition to long-term patterns, the weekly histogram shows sharp declines that exactly match recorded border interruptions in Gaza. For example, average supply shipments of trucks fell from approximately 1600 to approximately 500 during the Kerem Shalom attack (May 5–12, 2024). Similarly, weekly supply decreased from approximately 580 to approximately 180 during the Gaza Supply cut off (October 6-, 2024). Strong statistical evidence that each incident momentarily stopped the humanitarian supply chain is provided by the visual annotation of these events on the histogram. We can objectively evaluate the aid corridor's fragility and promote more robust logistical planning by measuring the percentage decline and recovery time.

Box Plot:

A box plot represents the distribution of data by showing the median, quartiles and outliers. It helps us to understand how the data is distributed. It is generally more suitable for numerical data rather than categorical data. Here, we have created a box plot of Quantity by Crossing, using the *boxplot()* function from the Seaborn library.



We can observe that, here, the Rafah Crossing has significantly higher variability and more extreme values. However, other crossings show relatively stable and consistent supply distribution. However, other crossings such as JLOTS and Kissufim show relatively shorter whiskers, which indicates more consistent and stable quantity distribution.

Chi-Square test of Independence:

The Chi-Square test of Independence is a non-parametric statistical method to determine whether there is a significant relationship between two nominal or categorical variables.

The steps to determine the association between two attributes are:

Either compare the chi-square value with the critical value; if the first one is greater than the other, then the null hypothesis (Ho) is rejected, else Ho proceeds, or compare the p-value;
if p < significant value, then Ho is rejected, else it holds.

We have used the *chi_contingency()* function of the scipy.stats module of Python for this test.

The followings are some performed test–

1. Checking that there is any relationship between Data Source and Description of Cargo?

```
# Checking that there is any relationship between Data Source and Description of Cargo?

# Null Hypothesis(Ho) : There is no relationship between Data Source and Description of Cargo

# Alternate Hypothesis(H1): There is a significant relationship between Data Source and Description of Cargo

from scipy.stats import chi2_contingency

# 1. Observation Table:

o_table = pd.crosstab(dataframe_copy["Data Source"],dataframe_copy["Description of Cargo"])

# 2. Expected values,chi_square value, Degrees of Freedom, p-values

chi2,p,df,expected = chi2_contingency(o_table)
```

```
# Values to watch

print(f"Chi_Square Value: {chi2}")

print(f"p Value: {p}")

# Decision

if p<0.05:

   print("Ho rejected. And there is a significant relationship between Data Source and Description of Cargo")

else:

   print("Ho accepted. And there is no significant relationship between Data Source and Description of Cargo")
```

Verdict→

Chi_Square Value: 108618.11296768082

p Value: 0.0

Ho rejected. And there is a significant relationship between *Data Source* and *Description of Cargo.*

2. Checking that there is any relationship between Data Source and the Crossing?

```
# Checking that there is any relationship between Data Source and the Crossing?

# Null Hypothesis(Ho): There is no relationship between Data Source and the Crossing

# Alternate Hypothesis(H1): There is a significant relationship between Data Source and the Crossing

from scipy.stats import chi2_contingency

# 1. Observation Table:

o_table = pd.crosstab(dataframe_copy["Data Source"],dataframe_copy["Crossing"])

# 2. Expected value, chi_square value, Degrees of Freedom, p-values

chi2,p,df,expected = chi2_contingency(o_table)

# Values to watch

print(f"Chi square value: {chi2}")

print(f"p value: {p}")

# Decision

if p<0.05:

   print("Ho rejected. And there is a significant relationship between Data Source and the Crossing")

else:
```

```
print("Ho accepted. And there is no significant relationship between Data Source and the Crossing")
```

Verdict→

Chi square value: 39063.472554122796

p value: 0.0

Ho rejected. And there is a significant relationship between *Data Source* and *the Crossing.*

3. Checking that is there any relationship between Description of Cargo and the Crossing?

```
# Checking that is there any relationship between Description of Cargo and the Crossing?

# Null Hypothesis(Ho) : There is no relationship between Description of Cargo and the Crossing

# Alternate Hypothesis(H1): There is a significant relationship between Description of Cargo and the Crossing

from scipy.stats import chi2_contingency

# 1. Observation Table:

o_table = pd.crosstab(dataframe_copy["Description of Cargo"],dataframe_copy["Crossing"])

# 2. Expected values,chi_square value, Degrees of Freedom, p-values

chi2,p,df,expected = chi2_contingency(o_table)
```

```
# Values to watch

print(f"Chi_Square Value: {chi2}")

print(f"p Value: {p}")

# Decision

if p<0.05:

   print("Ho rejected. And there is a significant relationship between Description of Cargo and the Crossing")

else:

   print("Ho accepted. And there is no significant relationship between Description of Cargo and the Crossing")
```

Verdict→

Chi_Square Value: 46562.571903666976

p Value: 0.0

Ho rejected. And there is a significant relationship between *Description of Cargo* and *the Crossing.*

Linear Regression:

Linear regression is a statistical model and machine learning algorithm used to model the linear relationship between a dependent variable (target) and one or more independent variables (features).

We have implemented a model to estimate the *quantity of the cargos* based on *the crossing locations*. In order to do this, we converted the category crossing data into numerical features using *one-hot encoding*.

This model was tested by the existing value of the data set; no new test data were involved.

It served as our first moves toward machine learning.

<u>Code</u>:

```
from sklearn.linear_model import LinearRegression

# Foucused on those datas which have pallets as unit

dataframe_copy=dataframe_copy[dataframe_copy["Units"]=="Pallets"
]

# Map Crossings (One-Hot Encoding). Additionally, creates X and
y variables

X = pd.get_dummies(dataframe_copy[['Crossing']],drop_first=True)

y = dataframe_copy['Quantity']

# Calculations

lr = LinearRegression()

lr.fit(X,y)
```

```
# Predictions

# Predicting for 'Kerem Shalom'

sample = X.head(1)

prediction = lr.predict(sample)



print(f"Predicted Quantity {prediction[0]} Pallets")
```

Verdict→

Predicted Quantity 22.77988284825188 Pallets

Conclusion:

To sum up, we as a team practiced a variety of statistical analyses that can be shown from this dataset. Moreover, we have learned about the infrastructure to send relief goods towards Gaza, Palestine. In addition, we are now able to assess which organizations have stronger ties to these kinds of contributions. In the final section, by developing the model, we can now guess from which crossing we may anticipate a precise amount of goods.

References:

[1] United Nations Relief and Works Agency for Palestine Refugees in the Near East (UNRWA), "State of Palestine: Gaza Supplies and Dispatch Tracking," *Humanitarian Data Exchange*, 2025. [Online]. Available: https://data.humdata.org/dataset/state-of-palestine-gaza-aid-truck-data. [Accessed: April 13, 2026].